

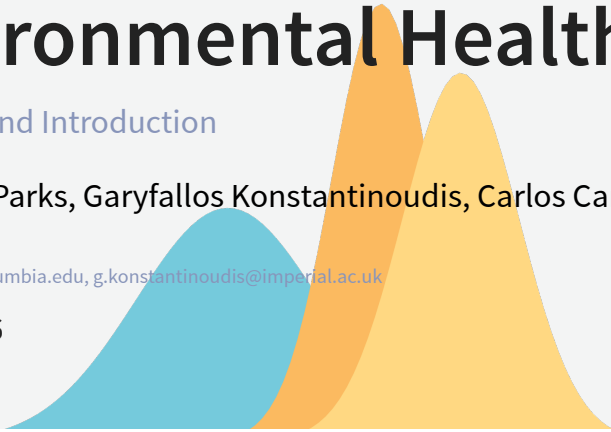
Bayesian Model(l)ing for Environmental Health

Welcome and Introduction

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Aug 5, 2026



Logistics

- **Wi-Fi Network:** guest-net (no password). Open any webpage on your browser, a pop-up will appear to connect to guest Wi-Fi. Accept terms to gain access.
- **Restrooms.** Directly outside the classroom to the right.
- **Course materials.** All material (lectures/labs) is located on Posit Cloud. You were invited to the 'sharp_bayesian_environmental_health_2026' workspace via email a few days ago.
- **Name tags.** Please wear during workshop to make connecting with others easier! Please return them after the Workshop to help us be greener.
- Contact Admin Coordinator Nader Elmehdawi ne2336@cumc.columbia.edu for any assistance.

Welcome!

Overview of workshop

The **Bayesian Modeling for Environmental Health Workshop** is a three-day intensive course of seminars and hands-on sessions to provide an *approachable* and *practical* overview of **concepts, techniques, and data analysis methods** used in Bayesian modeling with applications in Environmental Health.

Overview of workshop

- By the end of the workshop, participants should be familiar with the following topics:
 - Principles of Bayesian statistics
 - Bayesian workflow
 - Hierarchical modeling
 - Priors
 - Regression with non-linear terms
 - Spatial and spatio-temporal modeling

Overview of workshop

- By the end of the workshop, participants should be familiar with the following topics:
 - Software options for Bayesian modelling
 - INLA and R-INLA
 - Advanced environmental modelling
 - Attribution
 - Scaling Bayesian models
 - How this could apply to your research

Bayesian Modelling Workshop Team



Robbie M. Parks
Course Co-Director
Columbia University



Garyfallos Konstantinouidis
Course Co-Director
Imperial College London



Carlos Carrillo-Gallegos
Workshop Fellow
Columbia University



Tristan Sweden
Admin Coordinator
Columbia University



Nader Elmehdawi
Admin Coordinator
Columbia University

Special thank you

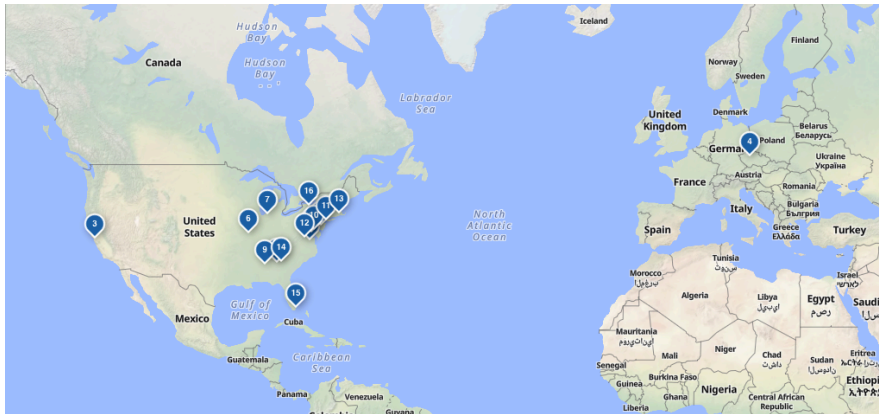


Theo Rashid
Instructor
Imperial College London



Elizaveta Semenova
Instructor
University of Oxford

12 US States + 1 country



Day 2 (August 6th 2026)

Time	Activity
8:30 - 9:00	Check in and breakfast
9:00 - 9:45	Regression with non-linear terms (Lecture)
9:45 - 10:30	Regression with non-linear terms (Lab)
10:30 - 10:45	Break
10:45 - 11:45	Spatial and spatio-temporal modeling (Lecture)
11:45 - 12:45	Lunch
12:45 - 1:45	Spatial and spatio-temporal modelling (Lab)
1:45 - 2:00	Break
2:00 - 2:30	Software options for Bayesian modelling (Lecture)
2:30 - 2:45	Break
2:45 - 3:45	INLA and R-INLA (Lecture)
3:45 - 4:30	INLA and R-INLA (Lab)
4:30 - 5:00	Questions, survey on projects, and wrap-up

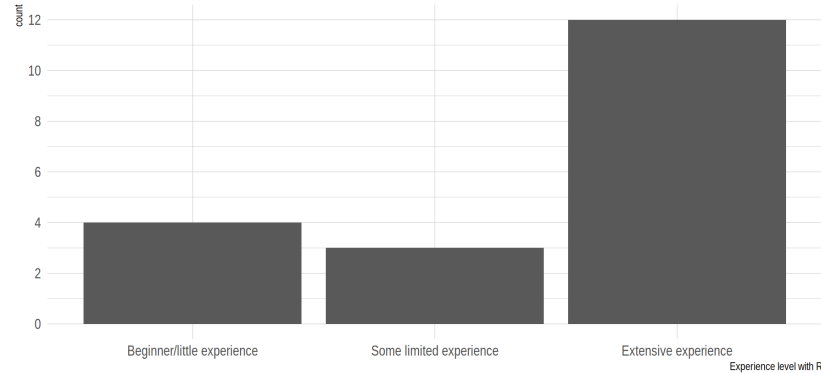
Day 1 (August 5th 2026)

Time	Activity
8:30 - 9:00	Check in and breakfast
9:00 - 9:15	Welcome and introduction
9:15 - 10:00	Principles of Bayesian statistics (Lecture)
10:00 - 10:45	Principles of Bayesian statistics (Hands-on Lab)
10:45 - 11:00	Break
11:00 - 11:45	Bayesian workflow (Lecture)
11:45 - 12:45	Lunch
12:45 - 1:30	Bayesian workflow (Lab)
1:30 - 1:45	Break
1:45 - 2:30	Hierarchical modeling (Lecture)
2:30 - 3:15	Hierarchical modelling (Lab)
3:15 - 3:30	Break
3:30 - 4:15	Priors (Lecture)
4:15 - 4:45	Priors (Lab)
4:45 - 5:00	Questions and wrap-up

Day 3 (August 7th 2026)

Time	Activity
8:30 - 9:00	Check in and breakfast
9:00 - 10:00	Advanced environmental modelling (Lecture)
10:00 - 10:15	Break
10:15 - 11:15	Advanced environmental modelling (Lab)
11:15 - 11:45	Attribution (Lecture)
11:45 - 12:45	Lunch
12:45 - 1:45	Scaling Bayesian models (Lecture)
1:45 - 2:15	Scaling Bayesian models (Lab)
2:15 - 2:30	Break
2:30 - 3:30	Potential projects discussion
3:30 - 4:00	Panel discussion
4:00 - 4:15	Group photo
4:15 - 5:00	Final farewell

What is your experience level with R?



Questions?

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